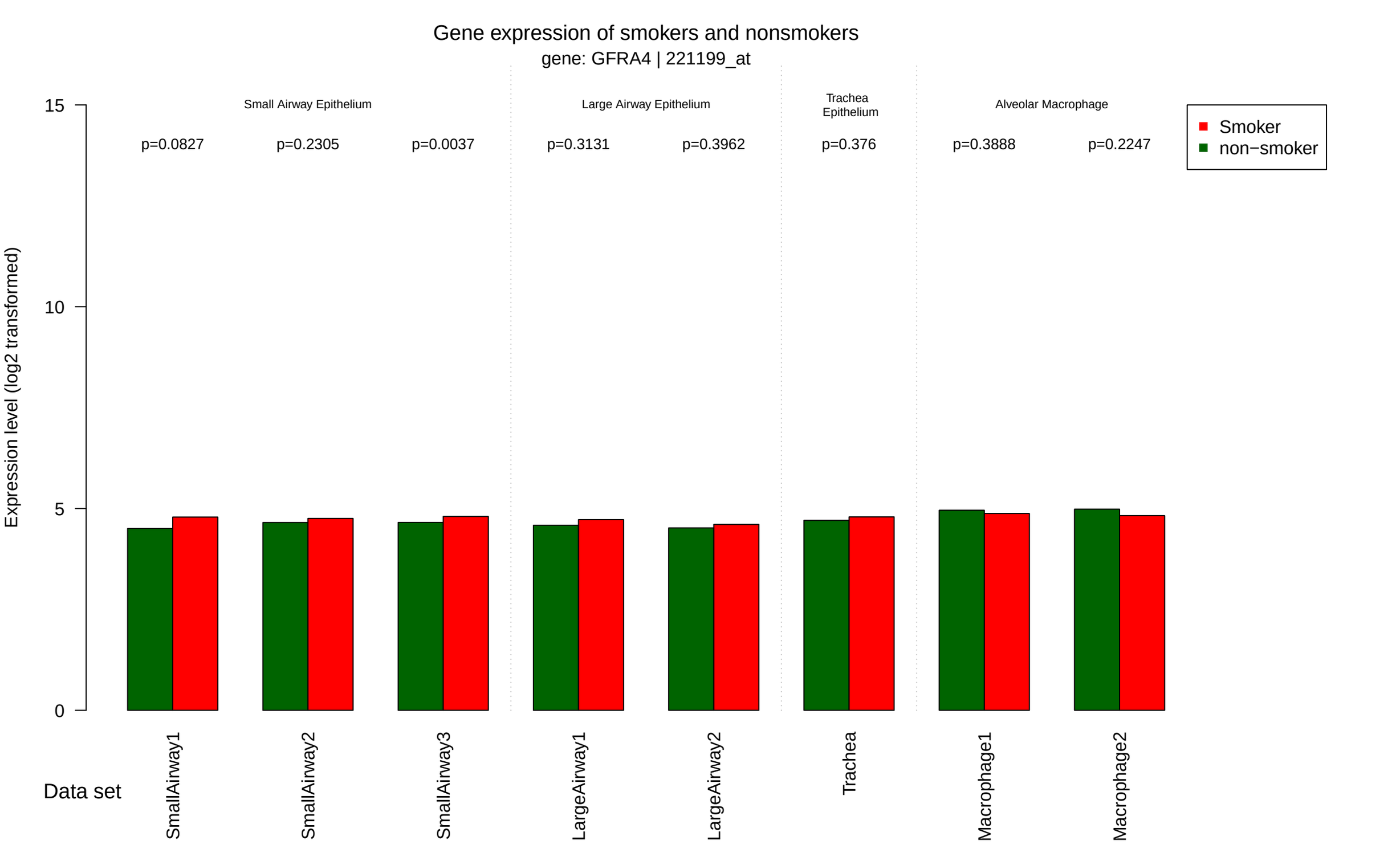
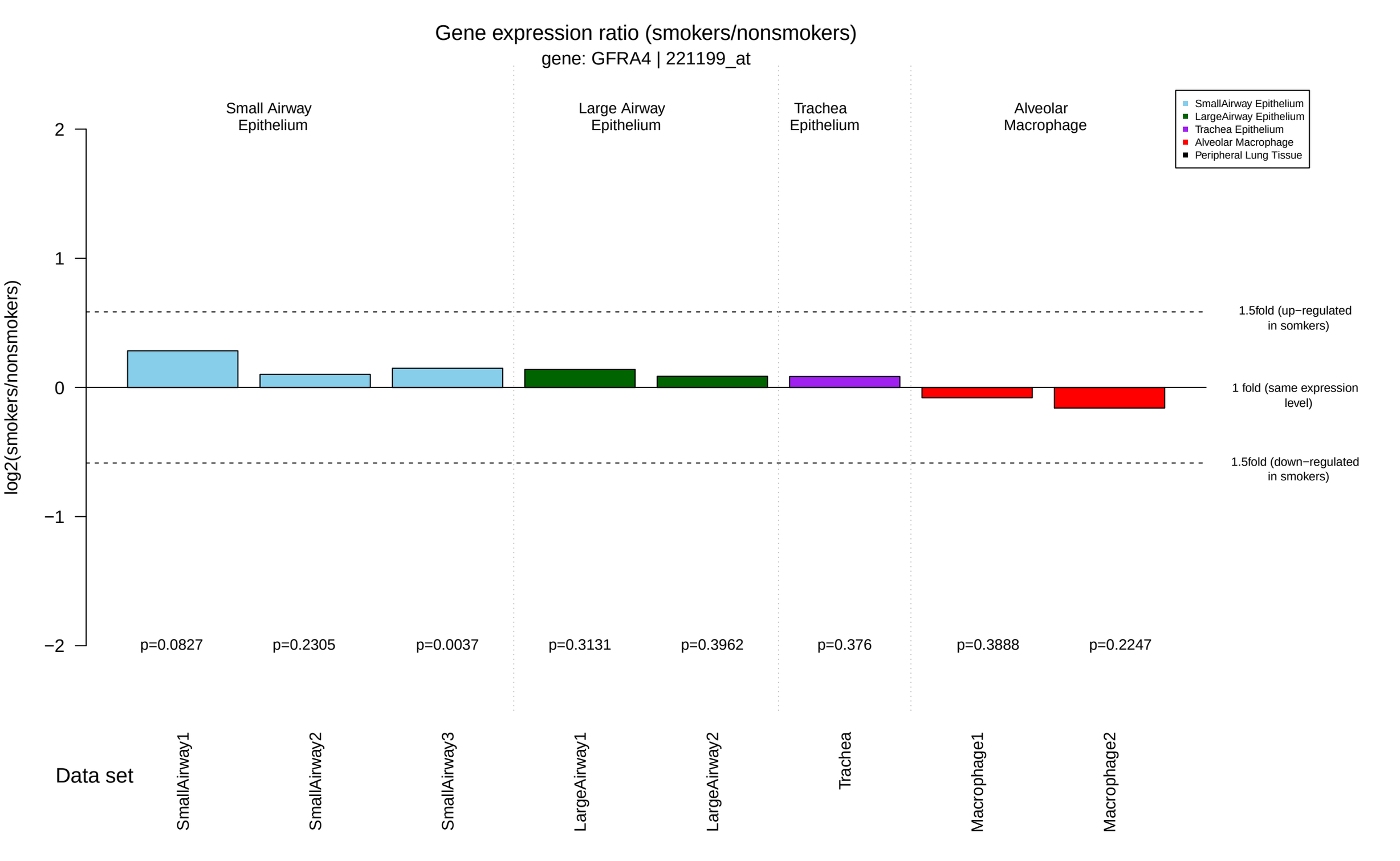
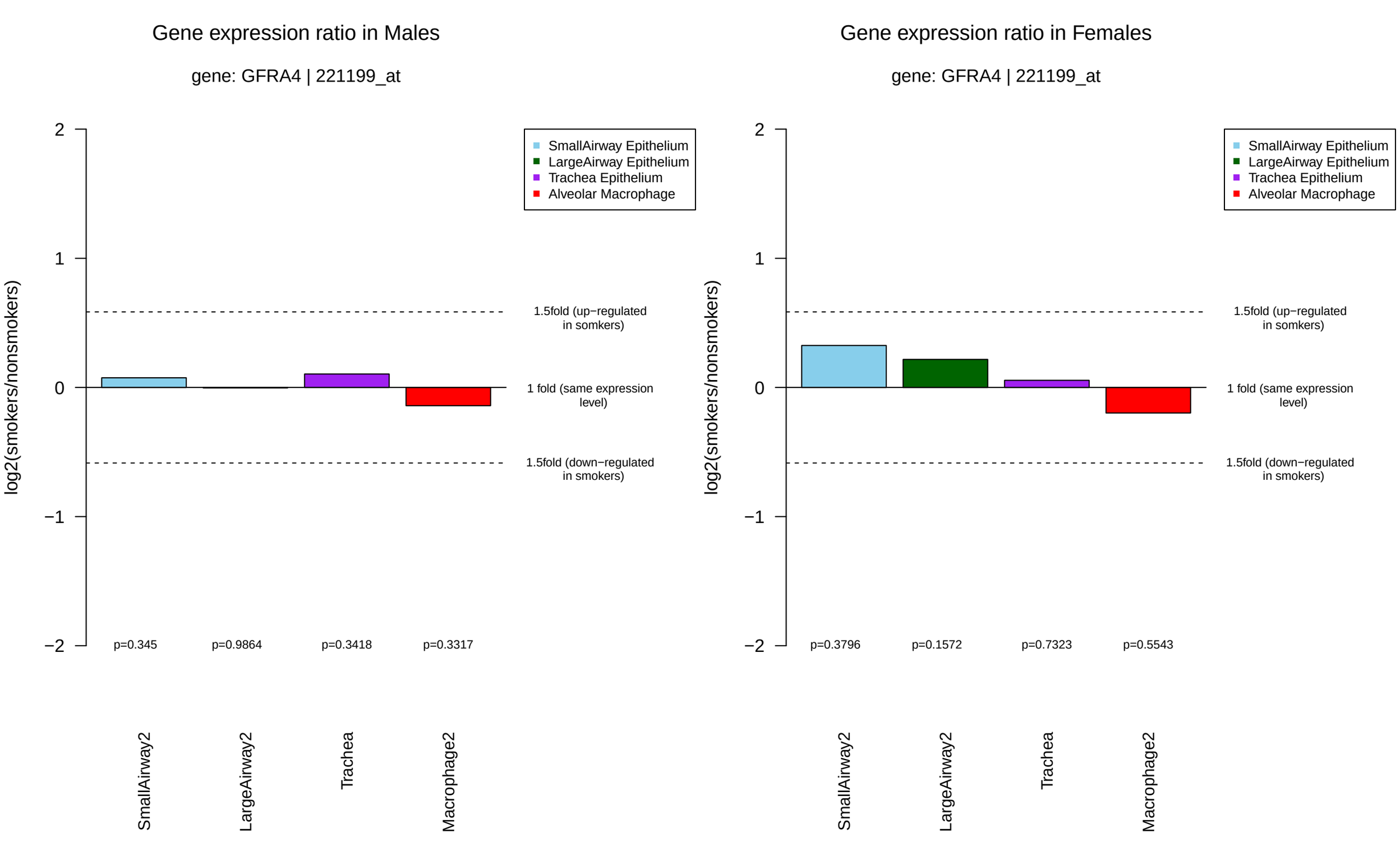
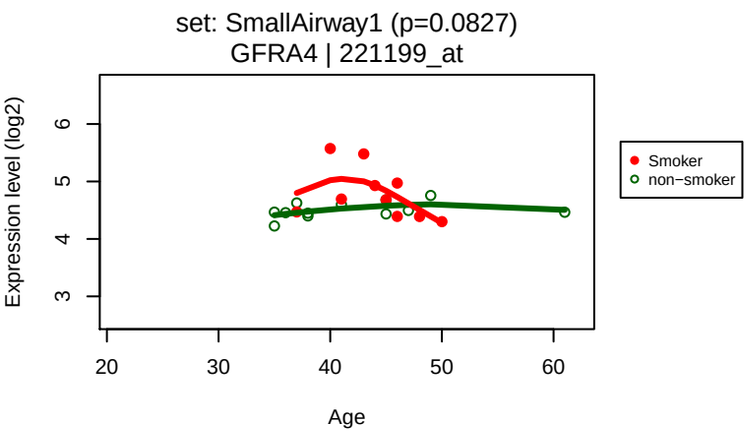
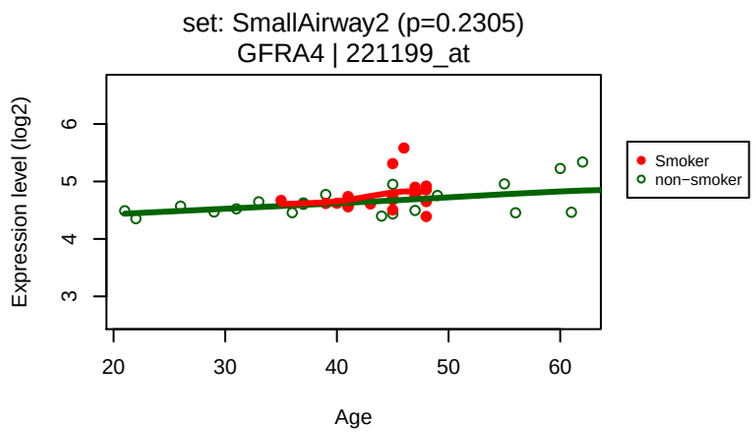
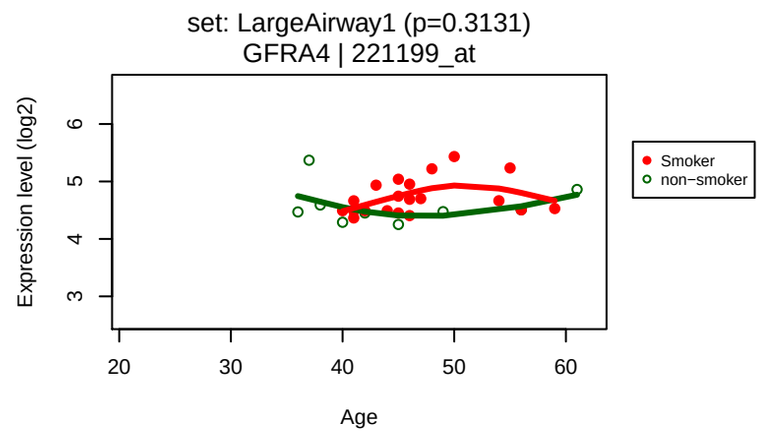

Gene expression of smokers and nonsmokers
gene: GFRA4 | 221199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0827
p=0.2305
p=0.0037
p=0.3131
p=0.3962
p=0.376
p=0.3888
p=0.2247
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

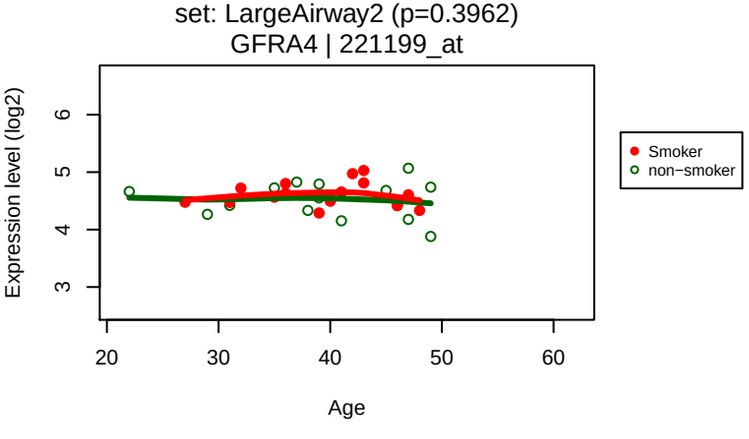

Gene expression ratio (smokers/nonsmokers)
gene: GFRA4 | 221199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0827
p=0.2305
p=0.0037
p=0.3131
p=0.3962
p=0.376
p=0.3888
p=0.2247
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

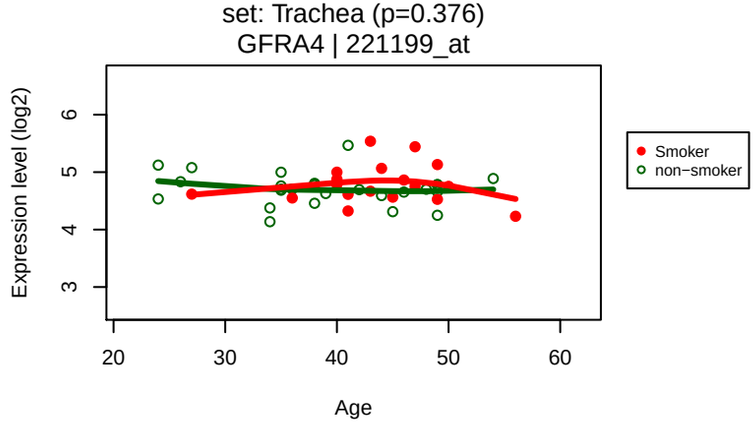

Gene expression ratio in Males
gene: GFRA4 | 221199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.345
p=0.9864
p=0.3418
p=0.3317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GFRA4 | 221199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3796
p=0.1572
p=0.7323
p=0.5543
SmallAirway2
LargeAirway2
Trachea
Macrophage2

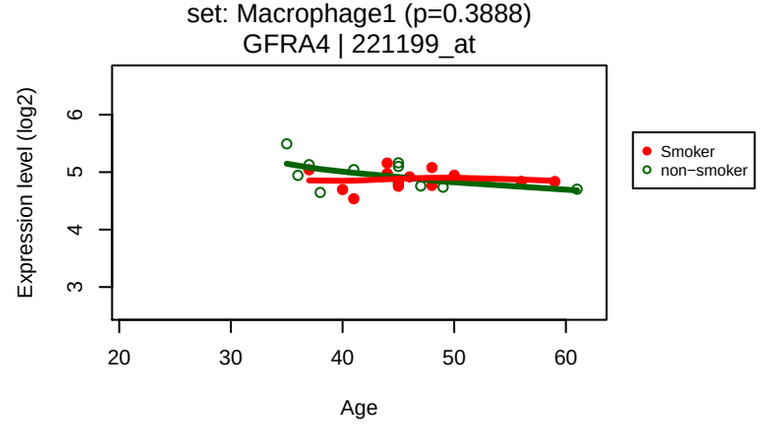

set: SmallAirway1 (p=0.0827)
GFRA4 | 221199_at
Expression level (log2)
Age
Smoker
non−smoker

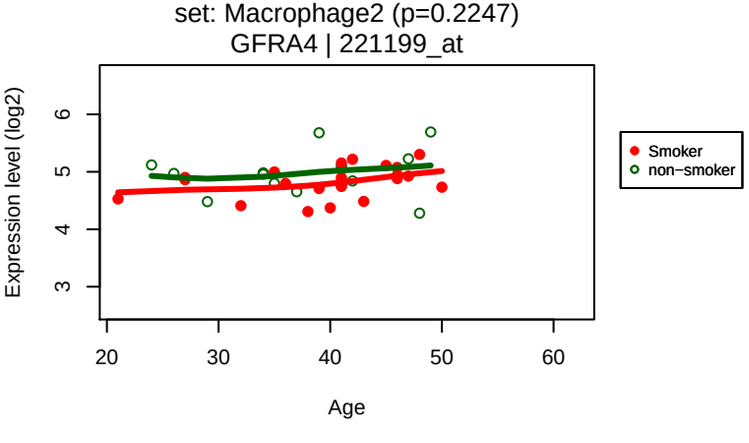

set: SmallAirway2 (p=0.2305)
GFRA4 | 221199_at
Expression level (log2)
Age
Smoker
non−smoker

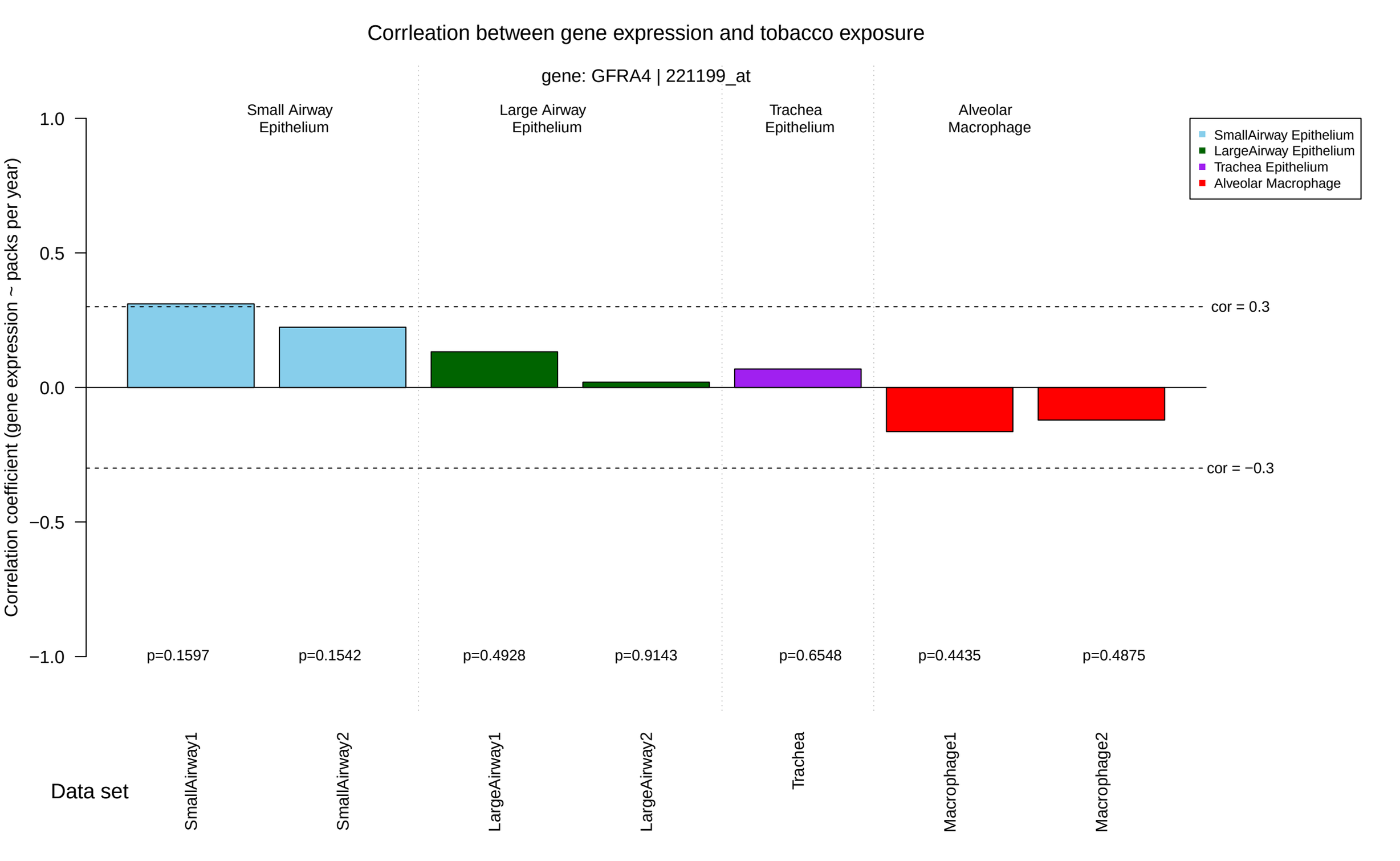

set: LargeAirway1 (p=0.3131)
GFRA4 | 221199_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3962)
GFRA4 | 221199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.376)
GFRA4 | 221199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3888)
GFRA4 | 221199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2247)
GFRA4 | 221199_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GFRA4 | 221199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1597
p=0.1542
p=0.4928
p=0.9143
p=0.6548
p=0.4435
p=0.4875
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2